

Candidate Name

Centre Number

Candidate Number



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

CHEMISTRY**9189/5**

PAPER 5 Practical Test

NOVEMBER 2014 SESSION

1 hour 20 minutes

Candidates answer on the question paper.
Additional materials:
as listed in Instructions to Supervisors.

TIME 1 hour 20 minutes**INSTRUCTIONS TO CANDIDATES**

Write your name, Centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

You are advised to show all working in calculations.

Use of a Data Booklet is unnecessary.

FOR EXAMINER'S USE

1	
2	
TOTAL	

This question paper consists of 8 printed pages.

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- 1 (a) You are required to investigate the kinetics of the reaction between FA1 and FA2.

The apparatus in Fig. 1 has been set up for you.

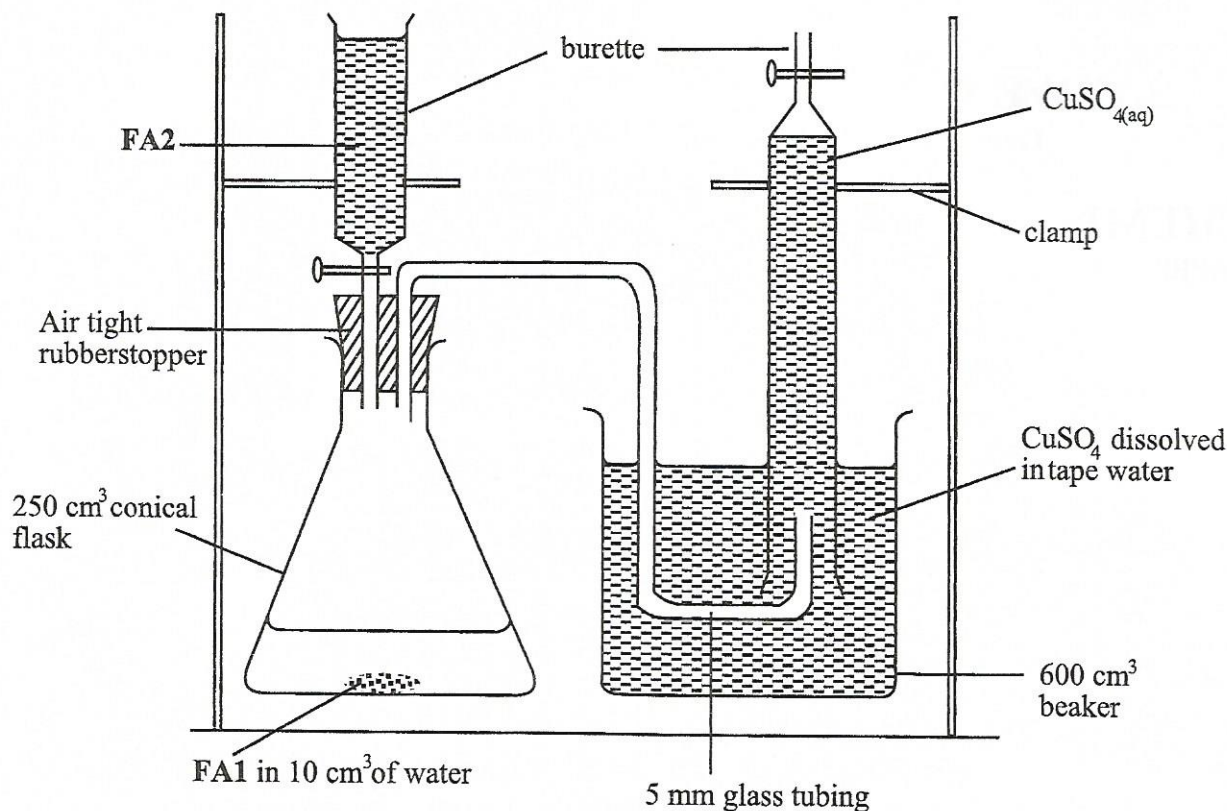


Fig. 1

FA1 is 0.02 g $\text{CuCO}_{3(s)}$ in 10 cm^3 of tap water

FA2 is 0.05 mol dm^{-3} $\text{HNO}_{3(aq)}$

- (i) Read and record the volume of copper sulphate in Table 1.

Start the reaction by releasing FA2 drop by drop into the conical flask and immediately start a stop watch.

- (ii) Read and record, in Table 1, the volume of copper sulphate at 1 minute intervals for 16 minutes.

Table 1.

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Time/min	0	1	2	3	4	5	6	7	8
Volume of $\text{CuSO}_{4(\text{aq})}/\text{cm}^3$									
Volume of gas/ cm^3									

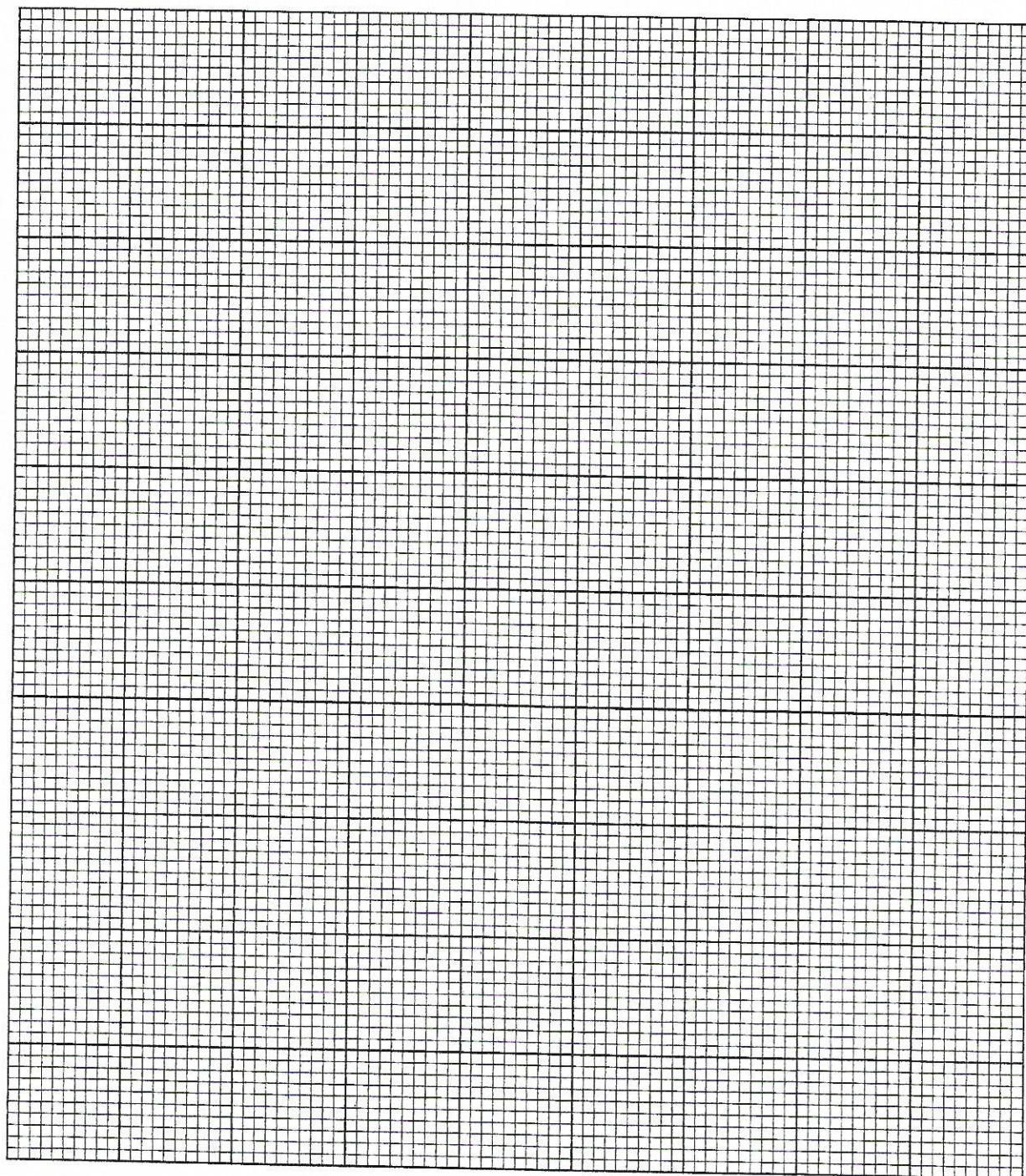
Time/min	9	10	11	12	13	14	15	16
Volume of $\text{CuSO}_{4(\text{aq})}/\text{cm}^3$								
Volume of gas/ cm^3								

(iii) Complete Table 1.

[8]

- (b) (i) Plot a graph of volume of gas produced against time.

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[3]

- (ii) Calculate the maximum speed of the reaction.

[3]

- (c) (i) Suggest **three** possible sources of error in this experiment.

1. _____

2. _____

3. _____

[3]

- (ii) Explain why

1. the copper sulphate was dissolved in tap water instead of distilled water.

2. 10 cm³ of tap water was added to FA1.

[2]

- (iii) State the purpose of the copper sulphate in this experiment.

[1]

- (d) (i) Write a balanced chemical equation for the reaction.

[1]

- (ii) Deduce the total volume of gas produced after 7 minutes.

[1]

- (iii) Calculate the mass of CuCO_3 that reacted after 7 minutes.

$$M_r[\text{CuCO}_{3(s)}] = 123.54$$

[2]

- (iv) Suggest how the results of this experiment can be made more reliable.

[1]

[Total: 25]

ASSESSMENT OF PLANNING SKILLS

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DO NOT CARRY OUT THE EXPERIMENT

- 2 You are required to plan and design an experiment to determine the enthalpy change of combustion of methane gas.

Assume you are provided with the following:

a small methane gas burner of mass 250 g
thermometer
tap water
250 cm³ beaker
burner stand
draught shield
a balance

- (a) Present your plan as a sequence of numbered steps.

The plan

[8]